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JOINERY ARRANGEMENT COMPRISING INTERLOCKED JOINERY COMPONENTS

Abstract

The disclosure relates to a joinery arrangement including a first and a second joinery component configured to be locked to each other. The first joinery component includes at least one cavity and a recess and the second joinery component includes at least one protrusion and an insert. The insert is displaceable and/or compressible along its main direction. The cavities and the protrusions are configured to cooperate with each other for locking the first and second joinery components to each other in a first direction and the recess and the insert are configured to cooperate with each other for locking the first and second joinery components to each other in a second direction. The insert includes a retaining member including at least one projection. The retaining member is configured to cooperate with the recess and/or an insert groove in a locked state.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to a joinery arrangement comprising joinery components configured to be locked to each other. The joinery arrangement may be included in a joinery assembly, such as a furniture assembly. The disclosure also relates to a joinery assembly comprising the joinery arrangement and additional joinery components.

BACKGROUND

[0002] WO 2023/191694 A1 discloses a locking device comprising a spring and a locking part, such as locking pin or a locking disc, for locking two elements, such as a first and a second panel, to one another. The first and/or second panels may be furniture components which may be a part of a furniture product. The locking pin may comprise an end part in the shape of a truncated cone, which may facilitate the locking of the two panels to one another.

[0003] Notwithstanding the advantages of the technology above, certain products, such as including furniture components, may require an even better locking performance.

SUMMARY

[0004] It is therefore an object of at least embodiments of the present disclosure to provide a joinery arrangement whose locking configuration is more resistant to high load, and especially to high impulse forces.

[0005] Another object of at least embodiments of the present disclosure is to provide such a joinery arrangement while maintaining the possibility to disassemble the joinery arrangement without damaging the joinery components.

[0006] Yet another object of at least embodiments of the present disclosure is to provide a corresponding joinery assembly comprising such a joinery arrangement.

[0007] These and other objects and advantages that will be apparent from the description have been achieved by the various aspects, embodiments and examples described below.

[0008] In accordance with a first aspect of the disclosure, there is provided a joinery arrangement comprising a first joinery component and a second joinery component configured to be locked to each other to obtain a locked state. The first joinery component comprises at least one cavity arranged in a mounting side of the first joinery component and extending along a cavity direction at a cavity angle with respect to a first plane extending along the mounting side of the first joinery component, and a recess arranged in the mounting side of the first joinery component. The second joinery component comprises at least one protrusion arranged in a mounting area of the second joinery component and extending along a protrusion direction at a protrusion angle with respect to a second plane extending along the mounting area of the second joinery component, and an insert arranged in the mounting area of the second joinery component, wherein the insert is displaceable and/or compressible along its main direction. The at least one cavity and the at least one protrusion are configured to cooperate with each other for locking the first and second joinery components to each other in a first direction perpendicular to said first plane. The recess and the insert are configured to cooperate with each other for locking the first and second joinery components to each other in a second direction along the first plane. The insert comprises a retaining member comprising at least one projection, such as a plurality of projections. The retaining member is configured to cooperate with a recess wall of the recess in the locked state. Alternatively, or additionally, the insert is arranged in the insert groove of the second joinery component, and the retaining member is configured to cooperate with a groove wall of the insert groove in the locked state.

[0009] By means of the retaining element in accordance with the present disclosure, higher friction between the insert and the recess and/or insert groove may be provided, such that the positioning of the insert in the recess and/or insert groove is better secured. For example, the insert may be prevented from sliding out of the recess and/or insert groove to a higher degree. Thereby, a higher locking strength may be obtained, and an unlocking of the joinery arrangement may be counteracted or even prevented when the first joinery component is subject to high load, and especially high impulse forces, along the first direction.

[0010] The joinery arrangement may be included in a joinery assembly, such as a furniture assembly. For example, the furniture assembly may be a cabinet assembly or a shelf assembly. It is emphasized that embodiments of the first and second aspects described herein are particularly suitable for a wall-mounted cabinet assembly, especially including a first joinery component forming a bottom, such as a bottommost, joinery component of the wall-mounted cabinet assembly.

[0011] The joinery components herein may be furniture components, such as cabinet components or shelf components. For example, the joinery components may be panels.

[0012] The joinery components herein may be wood-based joinery components, preferably comprising a core and, optionally, a décor structure, such as a décor layer attached, such as adhered, to the core. For example, the core may be a high-density fibre, HDF, board, a medium-density fibre, MDF, board, a particle board, or a plywood board. Moreover, the décor structure may be or may comprise a décor layer, such as a foil, a paper, a polymer-based layer, a laminated layer, or a wood veneer. Alternatively, or additionally, the décor structure may comprise a painted surface layer. The cavities, the recess and the insert groove (defined below) may be arranged in the core.

[0013] The cooperation, such as engagement, between the cavities and the protrusions as well as between the recess and the insert may define a locked state of the first and second joinery components. Furthermore, an unlocked state of the joinery components may be obtained when there is no cooperation between the cavity and the protrusion and/or the recess and the insert.

[0014] A transverse direction of the insert may be perpendicular to the main direction.

[0015] For simplicity of the presentation herein, reference will often be made to “projections” (or “cavities” or “protrusions”) instead of “at least one projection (or cavity or protrusion)”, but it is emphasized that “at least one projection (or cavity or protrusion)”, “at least two projections (or cavities or protrusions)” and “a plurality of projections (or cavities or protrusions)” are included in the concept of “projections” (or “cavities” or “protrusions”).

[0016] Generally herein, cooperating objects, such as surfaces, portions, members or elements, may cooperate by direct engagement or indirectly. Examples of cooperating objects include the cavities and protrusions, and the recess and the insert. In any embodiment described herein, there may be a sealing agent, such as a wax or an adhesive, provided between the cooperating objects, thereby providing examples of an indirect engagement.

[0017] The retaining member may be a friction element for increasing friction between the insert and the recess.

[0018] The projections may be separated from each other along the main direction of the insert by a spacing. For example, the projections may form corrugations on the insert. Thereby, a localized pressure from the retaining member against the recess may be provided.

[0019] In some embodiments, the projections may be evenly spaced. This may be advantageous when the first joinery component comprises a high-density board, such as high-pressure laminate, HPL, board, and/or a substantially homogeneous board.

[0020] The level of friction may be further fine-tuned to fit the material configuration, such as density profile, of the joinery components by adapting the spacings between the projections. Generally, the spacing may vary between different adjacent pairs of projections.

[0021] In some embodiments, the spacing between a first pair of adjacent projections may be smaller than the spacing between a second pair of adjacent projections, wherein the first pair is arranged closer than the second pair to an end section of the insert. Smaller and larger spacings

between the projections may provide a lower and higher pressure (and/or force) and friction against the core of the joinery component, respectively. Hence, a higher pressure (and/or force) and higher friction between the insert and the recess may be provided in an outer region of the joinery component than in an inner region of the joinery component arranged inside of the outer region along a thickness direction of the joinery component. For example, a wood-based first joinery component, preferably comprising a core in the form of an HDF board, an MDF board, a particle board, or a plywood board, and a décor structure, for example a décor layer comprising a polymer-based layer, a laminated layer or a wood veneer, may comprise a higher density in the outer region than in the inner region. Thereby, varying spacings may be particularly suitable for use together with such wood-based first joinery components since higher friction may be provided in the high-density region.

[0022] Another effect of the smaller spacing between the adjacent projections is that an adjusted pressure profile (and/or force profile) may be obtained, such that a sufficient locking strength may be provided while maintaining the possibility to disassemble the joinery components without damaging them. This may be particularly important for a wood-based first joinery component comprising a varying density profile as described above, for example comprising a core in the form of an HDF board, an MDF board, a particle board, or a plywood board, and a décor structure as described above.

[0023] In some embodiments, the projections may be separated from each other along a circumferential direction of the insert by a spacing.

[0024] The projections may extend around an entire circumference of the retaining member. Thereby, a simpler positioning of the insert into an insert groove may be provided. Additionally, at least in some applications, the manufacturing of the insert, such as by injection moulding, may become simpler.

[0025] The retaining member may generally taper in a longitudinal direction towards an end section of the insert. For example, transverse extensions of the projections may decrease towards the end section. Alternatively, or additionally, the recess wall may taper in a depth direction of the recess. For example, an end portion of the insert and/or the recess wall may have a frustoconical shape. Thereby, a larger contact area between the insert comprising projections and the recess may be provided, especially for a less dense core material, for example a particle board, whereby a higher locking strength along the first direction may be obtained. In addition, the wear of the recess may be reduced during disassembly of the joining arrangement.

[0026] Respective ones of the projections may extend, from a base surface of the retaining member, by different heights.

[0027] At least some of the projections may extend, from a base surface of the retaining member, up to a virtual frustoconical surface, which forms a non-zero angle relative to the base surface.

[0028] Transverse extensions of the projections may decrease along the main direction of the insert in a direction towards an end section of the insert.

[0029] In the locked state, a locking portion of the insert, such as of the retaining member, may cooperate with a locking surface of the recess and, optionally, there may be a play between a head surface of a head portion of the insert and a recess wall, preferably arranged opposite to the cooperating recess wall of the recess.

[0030] The mounting side of the first joinery component may be provided in a first face of the first joinery component. The mounting area may be provided in a first side edge of the second joinery component.

[0031] During locking, the insert may be configured to be compressed such that it assumes a compressed state, and, subsequently, the insert may be at least partially expanded towards an expanded state for locking the second joinery component to the first joinery component. For example, the insert may lock against the recess by pretension.

[0032] The second joinery component may be configured to be locked to the first joinery

component by relatively displacing the second joinery component with respect to the first joinery component in a locking direction extending essentially in parallel with at least one of the cavity direction and the protrusion direction.

[0033] The insert may be separately formed from the second joinery component. For example, the insert, including the retaining member and/or a body portion and serpentine elements (defined below), may be formed by injection moulding.

[0034] The insert may be arranged in an insert groove of the second joinery component.

[0035] The insert and/or an insert groove in which the insert is arranged may essentially extend along a normal direction of the second plane. Thereby, it may become easier to form the insert groove, such as by drilling.

[0036] The retaining member may comprise an outer and/or an inner retaining member. The outer (or inner) retaining member may be configured to cooperate with the recess (or the insert groove) in the locked state. It is emphasized that in some embodiments, the retaining member may comprise only an inner retaining member.

[0037] The at least one projection may be arranged in a body portion of the insert, wherein the body portion is configured to cooperate with a groove wall of the insert groove in the locked state. Thereby, a higher locking strength may be provided, and an unlocking of the joinery arrangement may be further counteracted or even prevented when the first joinery component is subject to high load, and especially high impulse forces, along the first direction. In addition, the insert may be better secured in the insert groove during said high load.

[0038] The insert may comprise a spring, such as a static load spring.

[0039] The insert may comprise a plurality of joined serpentine elements. The serpentine elements may be flexible and may implement a compressibility of the insert along the main direction, such as axial direction.

[0040] The protrusion angle may be smaller than the cavity angle in an unlocked state of the first and the second joinery components. Thereby, the first and second joinery components may be locked to each other by pretension. The protrusion angle may differ from the cavity angle by 0.5° to 6° , such as 2° to 4° , in the unlocked state. For example, the protrusion angle may be smaller than the cavity angle as described above.

[0041] Generally herein, the cavity angle and/or the protrusion angle may be acute, for example with respect to the first and/or the second plane. In non-limiting examples, the cavity angle may be between 20° and 75° , such as 30° to 60° , for example about 45° and/or the protrusion angle may be between 20° and 75° , such as 30° to 60° , for example about 42° .

[0042] The protrusion angle and the cavity angle may generally herein be measured in the unlocked state and in the locked state by a CT scanner.

[0043] Opposing walls of each cavity may be substantially parallel. A cross-section of the cavity and/or the recess may be circular.

[0044] In accordance with a second aspect of the disclosure, there is provided a joinery assembly comprising the joinery arrangement according to any of the embodiments of the first aspect and further comprising at least one additional joinery component, such as a plurality of additional joinery components. Embodiments of the second aspect are largely analogous to those of the first aspect, especially embodiments of the joinery arrangement, whereby reference is made thereto.

[0045] The joinery assembly may be a furniture assembly, such as a cabinet assembly or a shelf assembly.

[0046] Generally, all terms used herein, such as in the claims, are to be interpreted according to their ordinary meaning in the technical field, unless explicitly defined otherwise. All references to “a/an/the [element, device, component, means, step, etc.]” are to be interpreted openly as referring to at least one instance of said element, device, component, means, step, etc., unless explicitly stated otherwise. Reference to one or a plurality of “at least one element”, etc., may shortly be referred to as “the element(s)”.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0047] The disclosure will in the following be described in connection to exemplary embodiments and in greater detail with reference to the appended exemplary drawings, wherein:

[0048] FIGS. 1a-1c illustrate in a top view (FIG. 1a) and a cross-sectional side view (FIG. 1b) along the section B-B in FIG. 1a an exemplary embodiment of a joinery arrangement during locking, and in a partially sectional perspective view (FIG. 1c) an exemplary embodiment of a joinery arrangement in an unlocked state.

[0049] FIGS. 2a-2c illustrate in a cross-sectional side view along the section B-B in FIG. 1a an exemplary embodiment of a joinery arrangement during locking (FIG. 2a), in a partially sectional perspective view an exemplary embodiment of a joinery arrangement in a locked state (FIG. 2b), and in a cross-sectional side view an exemplary embodiment of a second joinery component without any insert and with one protrusion removed (FIG. 2c).

[0050] FIG. 2d illustrates in front views an exemplary embodiment of an insert in an expanded state (left subfigure) and in a compressed state (right subfigure).

[0051] FIGS. 3a-3b illustrate in a cross-sectional side view (FIG. 3a) along the section B-B in FIG. 1a and a corresponding enlarged cross-sectional side view (FIG. 3b) an exemplary embodiment of a joinery arrangement in a locked state.

[0052] FIGS. 4a-4c illustrate in a front view (FIG. 4a), a side view (FIG. 4b) and a perspective view (FIG. 4c) an exemplary embodiment of an insert in an expanded state.

[0053] FIGS. 4d-4f illustrate in a front view (FIG. 4d), a side view (FIG. 4e) and a perspective view (FIG. 4f) an exemplary embodiment of an end portion of an insert in an expanded state, which, e.g., may replace the end portions in FIGS. 4a-4c.

[0054] FIGS. 5a-5c illustrate in a perspective view (FIG. 5a) and in an enlarged front view (FIG. 5b) and an enlarged side view (FIG. 5c) an exemplary embodiment of an insert in an expanded state.

[0055] FIGS. 5d-5f illustrate in a perspective view (FIG. 5d) and in an enlarged front view (FIG. 5e) and an enlarged side view (FIG. 5f) an exemplary embodiment of an insert in an expanded state.

[0056] FIGS. 6a-6c illustrate in (enlarged) cross-sectional side views along a section B-B similar to that in FIG. 1a exemplary embodiments of a joinery arrangement in a locked state.

[0057] FIGS. 6d-6f illustrate in a perspective view (FIG. 6d) and in an enlarged front view (FIG. 6e) an exemplary embodiment of an insert in an expanded state, and in an enlarged side view (FIG. 6f) an exemplary embodiment of a joinery arrangement using such an insert in a locked state, corresponding to FIG. 6c.

[0058] FIGS. 7a-7b illustrate in an enlarged front view (FIG. 7a) an exemplary embodiment of an insert, and in an enlarged side view (FIG. 7b) an exemplary embodiment of a joinery arrangement using such an insert in a locked state.

[0059] FIGS. 7c-7e illustrate in a perspective view (FIG. 7c) and in an enlarged front view (FIG. 7d) an exemplary embodiment of an insert, and in an enlarged side view (FIG. 7e) an exemplary embodiment of a joinery arrangement using such an insert in a locked state.

[0060] FIGS. 7f-7h illustrate in an enlarged perspective view (FIG. 7f) and in an enlarged front view (FIG. 7g) an exemplary embodiment of an insert, and in an enlarged side view (FIG. 7h) an exemplary embodiment of a joinery arrangement using such an insert in a locked state.

[0061] FIGS. 8a-8c illustrate in a perspective view (FIG. 8a) and in an enlarged front view (FIG. 8b) an exemplary embodiment of an insert, and in an enlarged side view (FIG. 8c) an exemplary embodiment of a joinery arrangement using such an insert in a locked state.

[0062] FIGS. 8d-8g illustrate in frontal (FIG. 8d) and rear (FIG. 8e) perspective views an

exemplary embodiment of an insert, and in enlarged side views an exemplary embodiment of a joinery arrangement using such an insert (FIG. 8f) in a locked state, and an alternative exemplary embodiment of a joinery arrangement using an insert (FIG. 8g) in a locked state.

[0063] FIGS. 8h-8j illustrate in a perspective view (FIG. 8h) and in an enlarged front view (FIG. 8i) an exemplary embodiment of an insert, and in an enlarged side view (FIG. 8j) an exemplary embodiment of a joinery arrangement using such an insert in a locked state.

[0064] FIGS. 9a-9b illustrate in a perspective view an exemplary embodiment of a joinery assembly during assembly (FIG. 9a) and in an assembled state (FIG. 9b), e.g., including any of the joinery arrangements and/or inserts in FIGS. 1a-1c, 2a-2d, 3a-3b, 4a-4f, 5a-5f, 6a-6f, 7a-7h and 8a-8j.

[0065] FIGS. 10a-10c illustrate a side view (FIG. 10a), a front view (FIG. 10b), and a top view (FIG. 10c), samples of a joinery arrangement and a rig used for testing them in an impact load testing method.

[0066] FIGS. 11a-11c illustrate in diagrams measured test results of a peak load testing method (FIG. 11a) and of the impact load testing method (FIGS. 11b-11c) in FIGS. 10a-10c for various samples of a joinery assembly.

DETAILED DESCRIPTION

[0067] Next, various embodiments of a joinery arrangement **10** and a corresponding joinery assembly **20** will be described with reference to, e.g., FIGS. 1a-1c, 2a-2d, 3a-3b, 4a-4f, 5a-5f, 6a-6f, 7a-7h, 8a-8j, 9a-9b and 10a-10c.

[0068] The disclosure relates to a joinery arrangement **10** comprising a first **1** and a second **2** joinery component. The joinery components **1**, **2** may be furniture components, such as cabinet components or shelf components. For example, the joinery components may be panels, such as furniture panels.

[0069] The first (or second) joinery component **1** (or **2**) may comprise a first face **1b** (or **2b**), a second face **1c** (or **2c**) opposite to the first face **1b** (or **2b**). Moreover, the first (or second) joinery component **1** (or **2**) may further comprise a pair of opposing first **1d** (or **2d**) and second **1e** (or **2e**) side edges and a pair of side portions **1f**, **1g** (or **2f**, **2g**) extending between the faces **1b**, **1c** (or **2b**, **2c**) and between the side edges **1d**, **1e** (or **2d**, **2e**).

[0070] The first **1** and/or the second **2** joinery components may be wood-based joinery components, preferably comprising a core **18** and, optionally, a décor structure **19**, such as a decor layer attached to the core **18**. For example, the core **18** may be an HDF board, an MDF board, a particle board, a HPL board, or a plywood board. Moreover, the décor layer may be or may comprise a foil, a paper, polymer-based layer, a laminated layer, or a wood veneer. For example, the polymer-based layer may comprise a thermosetting resin, such as a melamine-formaldehyde resin, or thermoplastic polymers, such as polyvinyl chloride, PVC. Alternatively, or additionally, the décor structure **19** may comprise a painted surface layer.

[0071] The first joinery component **1** comprises at least one cavity **3**, such as a plurality of cavities **3**, and a recess **4**. The cavities **3** are arranged in a mounting side **1a** of the first joinery component **1** and extend along a cavity direction **P1** at a, preferably acute, cavity angle α with respect to a first plane **M1** extending along the mounting side **1a**. The mounting side **1a** may be provided in the first face **1b**. The recess **4** is arranged in the mounting side **1a**. The cavity angle α may be between 20° and 75°, such as 30° to 60°, for example about 45°. The second joinery component **2** comprises at least one protrusion **5**, such as a plurality of protrusions **5**, and an insert **6**. The protrusions **5** are arranged in a mounting area **2a** of the second joinery component **2** and extend along a protrusion direction **P2** at a, preferably acute, protrusion angle β with respect to a second plane **M2** extending along the mounting area **2a**. The mounting area **2a** may be provided in the first side edge **2d**. The protrusion angle β may be between 20° and 75°, such as 30° to 60°, for example about 42°.

[0072] Each protrusion **5** may be axially rigid. Thereby, it may be substantially incompressible along its axial direction. Each protrusion **5** may be a rod-shaped element, such as a cylinder-shaped

element. The protrusions **5** may be arranged in a hole **5c**, preferably by means of a press fit and/or an adhesive. The holes **5c** may be formed by drilling and may be inclined by an angle corresponding to the protrusion angle β . A cross-section of each protrusion **5** along the protrusion direction **P2** preferably is circular. Preferably, the protrusions **5** are separately formed from the second joinery component member **2**. For example, they may be plugs or dowels, preferably polymer-based dowels or wood dowels.

[0073] The insert **6** is arranged in the mounting area **2a** and is preferably arranged in an insert groove **9** of the second joinery component **2**. The insert **6** is displaceable and/or compressible along its main direction **A**, such as by being spring loaded. The main direction **A** may be an axial direction of the insert.

[0074] **A**, preferably maximal, length **K** of the protrusions **5** and/or a depth **L** of the cavities **3** may be 20-60 mm, such as 30-40 mm. **A**, preferably maximal, width **W1**, **W3** of the cavities **3** and/or protrusions **5** may be 5-10 mm, preferably about 8 mm. **A**, preferably maximal, width **W4**, **W5** of the insert **6** and/or insert groove **9** may be 5-12 mm, such as 5-10 mm or 8-12 mm. Likewise, a preferably maximal, width **W2** of the recess **4** may be 5-12 mm, such as 7-11 mm or 8-12 mm.

[0075] The cavities **3** and protrusions **5** are configured to cooperate, such as engage, with each other for locking the first **1** and second **2** joinery components to each other in a first direction **D1** perpendicular to the first plane **M1**. Each protrusion **5** may be received in a respective cavity **3** for cooperation therebetween. Moreover, the recess **4** and the insert **6** are configured to cooperate, such as engage, with each other for locking the first **1** and second **2** joinery components in a second direction **D2** along the first plane **M1**. Thereby, a locked state of the first **1** and second **2** joinery components in the first **D1** and second **D2** directions may be obtained, see, e.g., FIGS. **2b**, **3a-3b**, **6a-6c**, **6f**, **7b**, **7e**, **7h**, **8c**, **8f-8g**, **8j** and **9b**. Preferably, the first **D1** and second **D2** directions are perpendicular to each other. Clearly, the cavities **3** and the protrusions **5** may optionally also contribute to the locking of the second joinery component **2** to the first joinery component **1** in the second direction **D2**. In addition, the cavities **3** and the protrusions **5** and/or the recess **4** and the insert **6** may contribute to the locking of the second joinery component **2** to the first joinery component **1** in a third direction **D3**, which is perpendicular to the first **D1** and second **D2** directions. The cavities **3** and protrusions **5** and the recess **4** and the insert **6** may provide a locking device of the joinery arrangement.

[0076] A third plane **M3** extending along the face **2b** and/or **2c** may be configured to be arranged perpendicularly to the first plane **M1** in the locked state.

[0077] The protrusions **5** may cooperate with the respective cavities **3** at connecting surfaces **3b**, **5b** which may be arranged in a wall **3a** of the cavities **3** and in a wall **5a** of the protrusions. The connecting surfaces **3b**, **5b** may extend around a portion of a circumference of the protrusions and cavities, such as around their entirety.

[0078] The insert **6** may comprise a head portion **6g**, which preferably is generally tapering. The insert **6** may comprise a locking portion **6h** configured to cooperate with a locking surface **4c** of the recess **4**, preferably arranged in a recess wall **4a**, for locking the joinery components **1**, **2** in the second direction **D2**, see, e.g., FIGS. **3b**, **6a**, **6c** and **10a**. Preferably, a transversely inner head portion **6j** (or an outer head portion **6i**) facing an oppositely arranged recess wall **4b** and arranged opposite to the locking portion **6h** is spaced from the oppositely arranged recess wall **4b** by a play **15**. The locking portion **6h** and the inner **6j** (or outer **6i**) head portion may be provided in the head portion **6g**. For example, in the locked state, the inner **6j** (or outer **6i**) head portion may be closer than the locking portion **6h** to the cavities **3** and/or protrusions **5** along the **D2** direction. For example, a head length **HL** of the head portion **6g** may be 3-7 mm, such as 4-6 mm.

[0079] In the locked state, the insert **6**, such as a topmost portion **6k** along the main direction **A**, may be spaced from a bottom wall **4d** of the recess **4**.

[0080] Preferably, the cavities **3** and/or the recess **4** are blind holes. Moreover, the cavities **3** and/or the recess **4** may be circumscribed holes, such that they have extensions **H1**, **H2**, **H1'**, **H2'** in the

mounting area **2a** along a pair of non-parallel, such as perpendicular directions that are substantially the same. For example, they may be essentially circular or oval holes, preferably formed by drilling. A cross-section of each cavity **3** along the cavity direction **P1** preferably is circular, preferably such that a cavity entrance into the mounting side **1a** is oval. Alternatively, or additionally, a cross-section of the recess **4**, such as along the first direction **D1**, preferably is circular. Optionally, a cavity inlet **3c** is wider than an inner cavity portion **3d**, e.g., being formed by a countersink drill bit.

[0081] As shown most clearly in FIGS. **1b-1c**, **2a-2c**, **3a**, **6c** and **9a**, the recess **4** may be arranged in an edge portion **11a** of the joinery component **1** and/or the insert **6** may be arranged in an edge section **11b** of the second joinery component **2**. For example, the recess **4** may be arranged outside of each cavity **3** along the second direction **D2** and the insert **6** may be arranged outside of each protrusion **5** along the second direction **D2**. Thereby, as shown in FIG. **3b**, an unlocking hole **22** in the first joinery component **1**, such as that shown and discussed in relation to FIG. **10** in WO 2024/019644 A1, e.g., on page 17, lines 1-24, communicating with the recess **4** and configured to provide access to the insert **6** may be more easily concealed, cf. FIGS. **9a-9b**. However, as shown in FIG. **6a**, in some embodiments the recess **4** may be arranged between a pair of cavities **3**. This is also shown and discussed in relation to, e.g., FIGS. 4-10 in WO 2024/019644 A1, which content hereby is explicitly incorporated by reference.

[0082] In some embodiments, the protrusion angle β may be smaller than the cavity angle α in an unlocked state of the joinery components **1**, **2**. For example, the protrusion angle β may differ from the cavity angle α by 0.5° to 6° , such as 2° to 4° , in the unlocked state.

[0083] In some embodiments, however, the protrusion angle β may be essentially the same as the cavity angle α , for example differing by less than 0.5° , such as being about 0° .

[0084] In preferred embodiments, the insert **6** is separately formed from the second joinery component **2**, for example being formed by injection moulding. The insert **6** may comprise a polymer-based material, such as thermoplastic polymers. For example, the polymer-based material may comprise polyoxymethylene, POM, or polypropylene, PP, optionally comprising a reinforcing filler, such as glass fibers.

[0085] The protrusions **5** and the insert **6** may extend at different angles β and γ with respect to the second plane **M2**. The insert **6** may extend essentially along a normal direction **N2** of the second plane **M2**, preferably such that an insert angle γ between the main, such as axial, direction **A** of the insert **6** and a direction along the second plane **M2**, such as along the second direction **D2**, is $75^\circ \leq \gamma \leq 105^\circ$, for example about 90° . In some embodiments, however, the insert angle γ may deviate from the normal direction **N2** by up to $\pm 15^\circ$, such as being $75^\circ \leq \gamma \leq 105^\circ$ or $80^\circ \leq \gamma \leq 100^\circ$. The insert groove **9** may extend in the same direction as the insert **6**, whereby reference is made to the above values.

[0086] As shown in, e.g., FIGS. **4a-4f**, **5a-5f**, **6d-6f**, **7a-7h** and **8a-8j**, the insert **6** and/or its body portion **16**, such as its enveloping surface, may have a substantially cylindrical shape, although other shapes are equally conceivable.

[0087] Generally herein, the insert **6** may be displaceable and/or compressible along its axial direction **A**, see the arrows **C1** and **C2** in FIGS. **2a**, **2d**, **3a** and **4a** illustrating an inward and outward displacement and/or (de-)compression, respectively. For example, the insert may be axially flexible along the axial direction **A**. The insert **6** may assume an expanded state **S1** and a compressed state **S2**. For example, the insert may be at least partly compressed, such as entirely uncompressed, in the expanded state **S1**. The insert **6** may be configured to return towards, such as to return entirely to, the expanded state **S1** when compressed by means of a biasing force. Thereby, it may function as a compression spring. An extension **F1**, **F2** of the insert **6** along the axial direction **A** may be smaller in the compressed state **S2** than in the expanded state **S1**. A, preferably maximal, extension **F1** (in the expanded state **S1**) may be 30-60 mm, such as 45-55 mm. A, preferably maximal, extension **F2** (in the compressed state **S2**) may be 25-55 mm, such as 40-50

mm. Alternatively, or additionally, the compression F1-F2 may generally herein be 3-8 mm, such as 4-6 mm.

[0088] The insert may comprise a spring, such as a static load spring. As seen most clearly in FIGS. 4a-4c, 5a, 5d, 6d, 7c, 8a, 8d, 8e and 8h, the insert 6 may comprise a flexible, compressible member 6b, preferably in the form of a plurality of joined serpentine elements 6c, which, for example, may be S-shaped elements. The serpentine elements 6c may be connected to the body portion 16 of the insert 6, preferably in an integral manner (i.e., one piece). At least a part, such as an entirety, of the body portion 16 may be configured to be arranged in the insert groove 9 in the locked state, preferably engaging with a groove wall 9a and/or 9b of the insert groove 9. Preferably, the serpentine elements 6c are flexible. Moreover, the insert 6 may be configured to return towards the expanded state S1 when compressed by means of a biasing force of the serpentine elements 6c. The serpentine elements may thereby comprise alternating crests 6d and troughs 6e arranged along the axial direction A.

[0089] Adjacent serpentine elements 6c located along the same transverse side of the insert 6 may be separated by a space 6f along the axial direction A, at least in the expanded state S1, but preferably also in the compressed state S2.

[0090] In preferred embodiments, a, preferably maximal, width W2 of the recess 4 may be larger than a preferably maximal, width W4, W5 of the insert 6 and/or insert groove 9. For example, the width W4 of the insert 6 may be determined at the location of the projection(s) 8 (defined further below). The widths W2, W4, W5 may be determined along the second direction D2. Thereby, the insert 6 and/or insert groove 9 may be offset with respect to the recess 4 in the locked state. For example, a centre line CL1 of the insert 6 and/or insert groove 9 may be spaced along the second direction D2 from a centre line CL2 of the recess 4, wherein the centre lines CL1, CL2 extend along the first direction D1.

[0091] The first 1 and second 2 joinery components may be locked to each other by relatively displacing them in a locking direction 17, which extends essentially in parallel with the cavity direction P1 and/or the protrusion direction P2, preferably in a final phase of the locking. With reference to the first joinery component 1, one component of the locking direction 17 may include the first direction D1 and one component of the locking direction 17 may extend along the first plane M1, for example along the second direction D2 as shown in, e.g., FIG. 2a. Preferably, the third plane M3 is essentially arranged perpendicularly to the first plane M1 during locking. The locking may be accomplished without using a separate assembling tool.

[0092] During locking, the insert 6 may be compressed such that it assumes the compressed state S2, and, subsequently, the insert 6 may be at least partially expanded towards the expanded state S1 for locking the joinery components 1, 2. For example, the insert 6 may be entirely expanded to the expanded state S1. The insert 6 may be compressed during locking by means of a cooperation between the insert 6 and the first joinery component 1, preferably the mounting side 1a, see, e.g., FIG. 2a.

[0093] The insert 6 may be displaced during locking into the insert groove 7, preferably by being compressed such that it assumes the compressed state S2. Subsequently, the insert 6 may be displaced out from the insert groove 7, such that it at least partially is expanded towards the expanded state S1, for example entirely assuming the expanded state S1.

[0094] The protrusions 5 may each be received within the respective cavity 3 for locking the joinery components 1, 2.

[0095] Generally herein, the joinery arrangement 10 may extend along a primary X, a secondary Y, and a tertiary Z direction, which are perpendicular to each other. At least in the locked state, the primary X and tertiary Z directions may extend along the first M1 and second M2 planes, and the secondary direction Y may extend along a thickness direction of the first joinery component 1 and along the third plane M3.

[0096] The cavities 3 may be inclined with respect to the second direction D2, which, for example,

may be parallel with the primary direction X. For example, the primary X and/or tertiary Z directions may be configured to extend in parallel with a subfloor **30** in which the joinery arrangement **10** or joinery assembly **20** is to be installed, cf. FIGS. **9a-9b**, although other configurations are equally conceivable.

[0097] In some embodiments, the, preferably outer portion of the, protrusions **5** may be inclined in a direction away from the insert groove **9**, see, e.g., FIGS. **1a-1c**, **2a-2c**, **3a** and **6c**. In some embodiments, the protrusions **5** may be inclined in a direction toward the insert groove **9**, see, e.g., FIG. **9a**. The inclinations in FIGS. **9a-9b** may provide an improved locking of the joinery arrangement by means of an inward force towards a mounting wall **32** resulting from a downwardly directed gravitational pull on the joinery assembly **20**, cf. the discussion below. In any of these scenarios, the cavities **3** may be inclined in correspondence with the inclinations of the protrusions **5**. In yet some embodiments, some protrusions **5** may be inclined in a direction away from the insert groove **9** and some protrusions **5** may be inclined in a direction toward the insert groove **9**, see, e.g., FIG. **6a**.

[0098] Generally herein, the joinery components **1**, **2**, such as the faces **1b**, **2b**, **1c**, **2c** and/or side edges **1d**, **2d**, **1e**, **2e** and/or side portions **1f**, **2f**, **1g**, **2g**, may be rectangular or square, although other shapes are equally conceivable. In non-restrictive examples, a thickness **T1**, **T2** of the joinery components **1**, **2**, such as along the secondary Y or tertiary Z direction, may be 10-32 mm, preferably 12-19 mm. Moreover, in non-restrictive examples, the dimensions of the faces **1b**, **2b**, **1c**, **1c** may be between 300 mm and 1000 mm, such as along both of the primary X and tertiary Z directions or both of the primary X and secondary Z directions.

[0099] The insert **6** comprises a retaining member **7**, which comprises at least one projection **8**, such as two projections **8** or even a plurality of projections **8**, such as three to ten projections or four to eight projections (herein, not including the end section). The retaining member **7**, preferably in the form of an outer retaining member **7'**, may be configured to cooperate with the recess wall **4a** in the locked state. At least one projection **8**, such as at least two projections **8**, may be configured to cooperate with the recess wall **4a** in the locked state (see, e.g., FIG. **3b**). Hence, the cooperating portion of the projections **8** may be part of the locking portion **6h** and the locking surface **4c** may be part of the cooperating recess wall **4a**.

[0100] In the locked state, the locking portion **6h** may cooperate with the locking surface **4c** and a head surface **8d** of the head portion **6g** and a recess wall **4b** may be separated by the play **15**. The recess wall **4b** may be arranged opposite to the cooperating recess wall **4a**, for example along the second direction **D2** in the locked state. The head surface **8d** may be arranged in the inner **6j** (or outer **6i**) head portion. Preferably, the recess wall **4a**, **4b** is planar.

[0101] The retaining member **7** may be arranged in an end portion **14** of the insert **6** configured to face outwards from the insert groove **9**. The retaining member **7** and/or the end portion **14** may extend from the body portion **16**, preferably in an integral manner. The projections **8** may be arranged along the main direction A between a, preferably disc-shaped, longitudinal end section **6a** and the body portion **16**. A longitudinal extension of the retaining member **7** may be parallel with the main direction A of the insert **6**. In some embodiments, the projections **8** may form corrugations on the insert **6**.

[0102] Generally herein, the projections **8** may extend transversely, such as radially, outwards from a base surface **7a** of the retaining member **7**. The base surface **7a** may include transversely innermost portions of the retaining member **8**. In non-limiting examples, a height **Q** of the projections **8**, preferably from the base surface **7a**, may be at least 0.3 mm, preferably 0.3-1.0 mm, such as 0.4-0.8 mm. Moreover, a, preferably maximal, thickness **G** of the projections **8** may be at least 0.1 mm, preferably 0.1-0.8 mm, such as 0.2-0.6 mm. Also, a projection angle α of a top surface **8e** of the projections **8** may be 100°-125°, such as 105°-115°, with respect to the main axis A.

[0103] In some embodiments, the height **Q** may be constant along the main direction A of the insert

6, see, for example, FIGS. **5a-5f**, **6d-6f**, **7a-7h** and **8h-8j**. In some embodiments, the respective ones of the projections **8** may extend, from the base surface **7a** by different heights **Q** along the main direction **A** of the insert **6**, see, for example, FIGS. **3b**, **4a-4f**, **6d-6f**, **7a-7h** and **8a-8c**. For example, the height **Q** may be larger in a direction towards the topmost portion **6k** than in a direction towards the body portion **16**. Generally herein, the height **Q** may be determined along a transverse (radial) direction **TD** of the insert **6**.

[0104] As shown in, e.g., FIGS. **4a-4f**, **5a-5c**, **6d-6e**, **7a-7h**, **8a-8c** and **8h-8j**, at least some, such as all, of the projections **8** may extend around an entire circumference of the retaining member **7**, preferably along a circumferential direction **C** of the insert **6**. For example, the projections **8** may be annularly shaped or disc shaped. In some embodiments, however, at least one or some, such as all, of the projections **8** may extend around a portion of the circumference of the retaining member **7**, see, e.g., FIGS. **5d-5f** and **8a-8g**. In non-limiting examples, and as shown, the projections **8** may be arranged in a raster pattern or a lattice pattern.

[0105] The projections **8** may be formed by the same material as the insert **6**, preferably in one piece therewith. The material may comprise a polymer-based material. The projections **8** may be rigid, preferably such that they are essentially nondeformable, especially during locking. For example, the material of the projections **8** may be more rigid than the material of the core **18** of the first **1** and/or the second **2** joinery components, for example being wood-based joinery components.

[0106] The projections **8** may be separated from each other along the main direction **A** by a spacing **U**, **U1**, **U2**. In some embodiments, and as illustrated in, e.g., FIGS. **4d-4f**, **5a-5f**, **6d-6f**, **7a-7h** and **8h-8j**, the projections **8** may be evenly spaced such that the spacing **U** may be the same between all adjacent pairs of projections **8**. In some embodiments, the spacing **U1**, **U2** may vary between different adjacent pairs of projections **8**. As illustrated in, e.g., FIGS. **4a-4c**, **6d-6f**, **7a-7h** and **8a-8c**, the spacing **U1** between a first pair **8a** of adjacent projections **8** may be smaller than the spacing **U2** between a second pair **8b** of adjacent projections **8**. The first pair **8a** may be arranged closer than the second pair **8b** to the end section **6a**. It is clear to the skilled artisan that there may be more than two pairs, such as a plurality, of adjacent projections **8** for which the spacings **U** become gradually smaller towards the end section **6a**.

[0107] In yet some embodiments, the projections **8** may be separated from each other along the circumferential direction **C** of the insert **6** by a spacing **V**, see, e.g., FIGS. **5d-5f** and **8d-8f**, which show the spacing **V** in conjunction with the spacing **U**. The circumferential direction **C** may be perpendicular to the main direction **A**.

[0108] In non-limiting examples, the spacings **U**, **U1**, **U2** and/or **V** may be 0.3-1.5 mm, such as 0.4-1.0 mm.

[0109] Transverse extensions **E** of the projections **8** may generally vary along the main direction **A**, such as decrease towards the end section **6a**. Preferably, the retaining member **7** generally tapers in a direction towards the end section **6a**. The end portion **14** may have a frustoconical shape. For example, an envelope **7b** (which is a virtual surface) of the transverse extensions **E** may be arranged along straight line, e.g., such that the total envelope rotated around the main direction **A** is arranged along a frustoconical surface, see FIG. **4b**. In some embodiments, and as shown in FIG. **4d**, the envelope **7b** of the transverse extensions **E** may be arranged along a concave curve, e.g., such that the total envelope is arranged along a curved truncated conical surface.

[0110] As shown in, e.g., FIGS. **4a-4c** and **5a-5f**, the transverse extension **E** of a topmost projection **8c**, for example arranged in the end section **6a**, may be smaller than the transverse extension **E** of the projections **8** arranged longitudinally inside of the topmost projection **8c** along the main direction **A**. In some embodiments, however, and as shown in FIGS. **4d-4f**, the topmost projection **8c** may have a larger transverse extension **E** than the projections **8** arranged longitudinally inside of the topmost projection **8c** along the main direction **A**.

[0111] Also, the recess wall **4a** preferably tapers in a depth direction **12** of the recess **4**. The depth direction **12** may extend perpendicularly to the first plane **M1**. For example, the recess wall **4a** may

have a frustoconical shape.

[0112] In some embodiments, at least some (e.g., all) of the projections **8** may extend, from the base surface **7a** of the retaining member **7**, up to a virtual frustoconical surface, which forms a non-zero angle ϕ relative to the base surface **7a**.

[0113] In non-restrictive examples, an inclination angle ϕ between the generally tapering retaining member **7**, such as the envelope **7b** of the transverse extensions **E** of the projections **8**, and the main direction **A** may be $5^\circ \leq \phi \leq 20^\circ$, such as about 10° . Alternatively, or additionally, in non-restrictive examples, a wall angle θ between the recess wall **4a** and a normal direction **N1** of the first plane **M1** may be $5^\circ \leq \theta \leq 20^\circ$, such as about 10° . Also, the base surface **7a** may extend at an angle ω , for example of 5° - 20° , against the main direction **A**. Thus, the angle ξ relative to the base surface **7a** may correspond to the difference $|\omega - \phi|$.

[0114] In some embodiments, however, the recess wall **4a** may extend along the first direction **D1**, preferably by having parallel recess walls **4a**, **4b**, see FIG. **6b**. For example, the recess **4** may be shaped as a cylinder and/or the insert **6** may be substantially cylinder-shaped. Such configurations are imaginable in any of the embodiments herein, for example in any of FIGS. **1a-1c**, **2a-2d**, **3a-3b**, **4a-4f**, **5a-5f**, **6a-6f**, **7a-7h**, **8a-8g** and **9a-9b**.

[0115] According to some embodiments, and as shown in, e.g., FIGS. **6c-6f**, **7a-7h** and **8a-8g**, the at least one projection **8**, such as a plurality of projections **8**, of the retaining member **7**, preferably in the form of an inner retaining member **7''**, may be arranged in the body portion **16**. In particular, the projection(s) **8** may be arranged in an axially outer segment **16a** of the body portion **16** which is configured to face and/or engage with the groove wall **9a**, **9b** of the insert groove **9** in the locked state. For example, the axially outer segment **16a** may be configured to be disposed proximate to the entrance of the insert groove **9** into the mounting area **2a**. The body portion **16** may cooperate, such as engage, with each of a first groove wall **9a** and an oppositely arranged second groove wall **9b**.

[0116] It is emphasized that any of the embodiments in FIGS. **7a-7b**, **7c-7e**, **7f-7h** or **8a-8c**, FIG. **8d** together with FIGS. **8e-8f** or FIG. **8g**, or FIGS. **8h-8j** may be used in joinery components **1**, **2** of the type shown in, e.g., FIGS. **1a-1c**, **2a-2c**, **3a-3b** and **6c**.

[0117] The projections **8** may be configured to cooperate with the at least one groove wall **9a**, **9b**, such as each of the oppositely arranged groove walls **9a**, **9b**, and optionally configured to cooperate with the locking surface **4c**, preferably arranged in the recess wall **4a**. In a first example, as shown in FIGS. **8g-8j**, the at least one projection **8** may be arranged in the body portion **16** only. Here, the head surface **8d** of the, preferably generally tapering, head portion **6g** may, for example, be substantially flat and/or smooth. Such an insert **6** may provide a smoother, such as gentler, unlocking of the joinery components **1**, **2**, while providing a more secure arrangement of the insert **6** in the insert groove **9**. It is noted that the insert **6** in FIG. **8d** may represent a frontal perspective view of the insert **6** in FIG. **8g**, where the corresponding rear perspective view analogous to that in FIG. **8e** does not comprise a projection **8** in the end portion **14**. In a second example, and as shown in FIGS. **6c-6f**, **7a-7h** and **8a-8f**, at least two, such as a plurality, projections **8** may be arranged in the body portion **16** as well as in the end portion **14** configured to cooperate with the recess **4**. As described elsewhere herein, the end portion **14** in which a part of the retaining member **7** comprising the projection(s) **8** is arranged, may generally taper.

[0118] The outer **7'** and/or inner **7''** retaining member may extend only around a portion of the circumference of the retaining member **7**, such as along the circumferential direction **C**, see, e.g., FIGS. **6d**, **8a** and **8d-8e**. Alternatively, the outer **7'** and/or inner **7''** retaining member may extend around an entire circumference of the retaining member **7**, see, e.g., FIGS. **6d**, **7c**, **7f**, **8a** and **8h**.

[0119] The projection(s) **8** arranged in the axially outer segment **16a** may extend transversely outside of an axially inner segment **16b** of the body portion **16**, see, e.g., FIGS. **6c-6f**, **7a-7b**, **7f-7h** and **8a-8g**. Alternatively, the projection(s) **8** arranged in the axially outer segment **16a** may extend to, or transversely inside of the axially inner segment **16b**, see, e.g., FIGS. **7c-7e** and **8h-8j**. The

transverse extensions may extend along the transverse direction TD of the insert **6**. For example, any of the relations above may be specified in a front view and/or a side view of the insert **6**. In any of these embodiments, the axially outer **16** and inner **16b** segments may be located on the same transverse side of the insert **6**.

[0120] Generally herein, the action of the retaining member **7** may increase when the load is increased. As illustrated in FIG. **6c**, when the first joinery component **1**, such as the first face **1b**, is loaded, a force *F* directed along the first direction **D1** may be created, such that the first joinery component **1** may be displaced relative to the second joinery component **2** in a direction antiparallel to the first direction **D1** and along the second direction **D2** due to the inclined protrusions **5** arranged in the inclined cavities **3**. Due to a resulting force *RF* from the recess **4** acting on the insert **6** along the second direction **D2**, the outer retaining member **7'** may become activated by means of an increased engagement strength with the recess wall **4a**. The force *RF* may also cause the insert **6** to be pressed against the insert groove **9**, such that an oppositely directed force *RF'* in the insert groove **9** may activate the inner retaining member **7''** by means of an increased engagement strength with the second groove wall **9b**. The activation of the outer **7'** and inner **7''** retaining members by means of the load may counteract an unlocking of the joinery components **1**, **2**.

[0121] The quantities specified herein, especially the angles α , β , γ , σ , ϕ , θ , ω , ξ or the extensions, *E*, **F1**, **F2**, *G*, **HL**, *K*, *L*, *Q*, **T1**, **T2**, *U*, **U1**, **U2**, *V*, **W1-W5**, may be determined by means of a profile projector, for example from Mitutoyo or Dravitec AB. The joint gap *J* may be measured with a digital dial gauge.

[0122] FIGS. **9a-9b** illustrate embodiments of a joinery assembly **20** comprising the joinery arrangement **10** and/or inserts **6** according to any of the embodiments shown in, e.g., FIGS. **1a-1c**, **2a-2d**, **3a-3b**, **4a-4f**, **5a-5f**, **6a-6f**, **7a-7h** and **8a-8j** and further comprising at least one additional joinery component **11**, **2**, **2'**, such as a plurality of additional joinery components. In fact, this particular embodiment includes two joinery arrangements **10** with two second joinery components **2** locked to the same first joinery component **1**. The mounting side **1a** may be an inner first face **1b** of the joinery assembly **20**. The joinery assembly **20** may be a furniture assembly. For example, the furniture assembly may be a cabinet assembly or a shelf assembly. FIG. **9b** illustrates a wall-mounted cabinet assembly comprising a first joinery component **1** which forms a bottom joinery component of the wall-mounted cabinet assembly, such as along the secondary direction *Y*. The wall-mounted cabinet assembly is arranged at a vertical distance **31** from the subfloor **30** and may be fixed to the mounting wall **32** by any means known to the skilled artisan, such as by screws and/or angle brackets.

[0123] By way of example, the additional joinery components **11**, **2**, **2'** may be selected from the group consisting of a side joinery component **2**, a top joinery component **1'**, and a backside joinery component **2'**. The additional joinery components **11**, **2**, **2'** may be connected to each other and/or to the joinery arrangement **10** by means of any embodiment described herein, such as in any of FIGS. **1a-1c**, **2a-2b** and **3a-3b**, or by any means known in the art, for example by screws or by means of inclined rod-shaped elements and insertion recesses as well as a compressible locking element and a recess, e.g., as disclosed in WO 2023/143756 A1. Optionally, one or several of the joinery components **1**, **11**, **2** may comprise a connecting channel **21** for connecting the backside joinery component **2'**.

[0124] In any embodiment herein, the end portion **14** may comprise an indicator **14a**, such as a notch, for indicating a rotational position of the insert **6** around the main axis *A*, cf. FIGS. **8h-8i**.

EXAMPLES

[0125] A reference set **RO** of reference samples and three sets **R1**, **R2**, **R3** comprising samples of a joinery arrangement or joinery assembly were tested in accordance with a peak load testing method and an impact load testing method. Each sample in each of the sets **R0**, **R1**, **R2**, **R3** comprised a first and a second joinery component in the form of panels with dimensions 130×16 mm×60 mm

and 180 mm×300 mm×16 mm along the X, Y, Z directions, respectively, and had a core in the form of a particle board and a decorative layer comprising melamine formaldehyde resin. The first joinery component comprised recess having a tapering frustoconically shaped recess wall with a wall angle θ of 10°. A maximal width W_2 , corresponding to a diameter of the recess, was 11.2 mm. The sets **R0**, **R1-R3** further comprised an insert consisting of pure POM without any filler and being designed identically, except for the head portions. The inserts had frustoconically shaped end portions and a maximal extension F_1 of about 51 mm, and each head portion had a minimal transverse extension E of 4 mm, a head length HL of 5.3 mm, and an inclination angle ϕ of 10°. The reference set **R0** comprised an insert without any projections, having a substantially smooth head portion in accordance with the embodiment in FIGS. 1a-1c of WO 2023/191694 A1, and having a surface roughness of 27 according to the standard VDI (Verein Deutscher Ingenieure). All projections in **R1-R3** had $G=0.3$ mm perpendicularly to the base surface. The first set **R1** comprised six evenly spaced projections (cf. FIGS. 5a-5c) with $U=0.8$ mm and $Q=0.7$ mm. The second set **R2** comprised five evenly spaced separated projections arranged along a concave curve (cf. FIGS. 4d-4f) with $U=0.8$ mm and $Q=0.7$ mm. The third set **R3** comprised four projections with spacings becoming gradually smaller towards the end section (cf. FIGS. 4a-4c) with $0.9 \text{ mm} \leq U \leq 1.3 \text{ mm}$, $0.5 \text{ mm} \leq Q \leq 1.0 \text{ mm}$, and the base surface extended at an angle $\omega=16^\circ$.

[0126] According to the peak load testing method, a strength of a locking device under an application of an excessive load on a sample of a bottom panel of a joinery assembly, such as being configured to be wall mounted, was estimated. The samples in the sets **R0**, **R1-R3**, were conditioned for the test at a temperature of $23 \pm 2^\circ \text{ C}$. and a relative humidity of $50 \pm 5\% \text{ RH}$ for a minimum of 24h. Two samples of side panels were assembled to the bottom panel similarly to the joinery components 1, 2 in FIGS. 9a-9b and a sample of a top panel was assembled to the side panels similarly to the top joinery component 1' in FIG. 9b (no backside panel was used). A hole was drilled in the top panel at a geometrical center so that a pressure bar could be freely inserted through the top panel. A load was applied on 100% of a depth and on 75% of a width (centered) of the bottom panel (corresponding to dimensions along the primary X and tertiary Z directions in, e.g., FIGS. 9a-9b). The resulting arithmetic mean static load in the form of a peak load at break was then measured for three samples of the joinery assembly in each of the sets **R0**, **R2** and **R3**, and the results are presented in FIG. 11a (see the hatched bars). It is clear from FIG. 11a that the samples in the second **R2** and third **R3** sets were resistant to a higher load as compared to the samples of the reference set **R0**.

[0127] Other samples in sets **R0'**, **R1'**, **R2'**, **R3'** were also tested in accordance with the peak load testing method and the impact load testing method. The samples of the sets **R0'**, **R1'**, **R2'**, **R3'** were identical to those in the sets **R0**, **R1**, **R2**, **R3**, respectively, except that the recess was cylindrically shaped, hence having a recess wall with a wall angle θ of about 0°. A maximal width W_2 , corresponding to a diameter of the recess, was 11 mm. Such a cylindrical recess thus had a smaller contact area than the conical recess.

[0128] The resulting arithmetic mean static load in the form of a peak load at break was then measured for three samples of the joinery assembly in each of the sets **R0'**, **R2'** and **R3'**, and the results are presented in FIG. 11a (see the solid bars). It may be deduced from FIG. 11a that the samples in the second set **R2'** (or third set **R3'**) were resistant to a higher (or lower) load than the samples of the reference set **R0'**. It may also be deduced that while the samples in the second sets **R2**, **R2'** could endure about the same load, the samples in the third set **R3** were resistant to a higher load than the samples of the third set **R3'**. In other words, the performance of the conical recess was better than that of the cylindrical recess.

[0129] According to the impact load testing method, and with reference to the drawings in FIGS. 10a-10c where an associated test rig is detailed, a joint gap J along the first direction D_1 (see FIG. 2a) of five assembled samples of each of the sets, **R0**, **R1-R3** arranged in the rig resulting from a dropping weight was measured by means of a digital dial gauge. The result of the testing method

indicated a mechanical endurance of the locking device under impact loads.

[0130] The method included the following steps for each assembled sample in the sets **R0**, **R1-R3** after being conditioned as described above. The sample was arranged in the rig and the joint gap **J** was measured. As shown in FIGS. **10a-10c**, a homogenous weight element **40** of the mass 0.740 kg in the form of a vertically aligned disc of diameter 80 mm and width 9.3 mm was repeatedly dropped in impact engagement with the first face of a side panel at a vertical distance of 100 mm from the sample of the bottom panel. The resulting joint gap **J** in the joint was measured after each impact, the arithmetic mean result of which is summarized in FIG. **11b**. The dropping of the weight was repeated until a released state was achieved, when the topmost portion of the insert was situated at or above the first face extending along the mounting side of the bottom panel, such as when the bottom panel broke and/or when a complete disengagement of the insert from the recess was achieved.

[0131] The arithmetic mean number of impacts for achieving the released state is summarized in FIG. **11c**.

[0132] In view of the above, it may be deduced that, for any given number of impacts, the resulting arithmetic mean joint gap **J** for the samples in the second set **R2** (or each of the first **R1** and third **R3** sets) was smaller (or larger) than that in the reference set **R0**. Hence, the samples of the second set **R2** (or the first **R1** and third **R3** sets) generated on average the highest (or lowest) friction between the insert and the recess. Furthermore, while the samples in the second **R2** and third **R3** sets on average endured a greater number of impacts than those in the reference set **R0**, the samples in the first set **R1** endured a fewer number of impacts.

[0133] In fact, the friction of the samples of the second set **R2** was found to be too high for some applications involving the particle board as a core, since an unlocking of some of these samples resulted in a breaking, such as chipping, of them. The samples were unlocked by means of an unlocking hole **22** as discussed above. On the other hand, the unlocking of the samples of the third set **R3** was found to be less destructive and typically did not result in any damage of the samples. The inventors have therefore surprisingly found that, at least for some applications, e.g., involving a particle board as a core, the samples of the third set **R3** may constitute a balanced compromise for providing a sufficiently high locking strength while maintaining the possibility to disassemble the joinery components with little or no damage.

[0134] Aspects of the disclosure have mainly been described above with reference to a few embodiments. However, as is readily appreciated by a person skilled in the art, other embodiments than the ones disclosed above are equally possible within the scope of the disclosure. For example, it is clear that in some embodiments the outer **6i** and/or inner **6j** head portion may cooperate with the recess wall **4b**, such that there is no play **15**.

Claims

1. A joinery arrangement comprising a first joinery component and a second joinery component configured to be locked to each other to obtain a locked state, wherein: the first joinery component comprises: at least one cavity arranged in a mounting side of the first joinery component and extending along a cavity direction at a cavity angle with respect to a first plane extending along the mounting side of the first joinery component, and a recess arranged in said mounting side of the first joinery component; and the second joinery component comprises: at least one protrusion arranged in a mounting area of the second joinery component and extending along a protrusion direction at a protrusion angle with respect to a second plane extending along the mounting area of the second joinery component, and an insert arranged in the mounting area of the second joinery component, the insert being displaceable and/or compressible along its main direction, wherein said at least one cavity and said at least one protrusion are configured to cooperate with each other for locking the first and second joinery components to each other in a first direction perpendicular to

said first plane, wherein the recess and the insert are configured to cooperate with each other for locking the first and second joinery components to each other in a second direction along said first plane, wherein the insert comprises a retaining member comprising at least one projection, such as a plurality of projections, and wherein the joinery arrangement is configured such that, in the locked state: the retaining member cooperates with a recess wall of the recess; and/or the insert is arranged in an insert groove of the second joinery component, and the retaining member cooperates with a groove wall of the insert groove.

2. The joinery arrangement according to claim 1, wherein the projections are separated from each other along the main direction of the insert by a spacing.

3. The joinery arrangement according to claim 2, wherein the spacing between a first pair of adjacent projections is smaller than the spacing between a second pair of adjacent projections, the first pair being arranged closer than the second pair to an end section of the insert.

4. The joinery arrangement according to claim 1, wherein the projections are separated from each other along a circumferential direction of the insert by a spacing.

5. The joinery arrangement according to claim 1, wherein the projections extend around an entire circumference of the retaining member.

6. The joinery arrangement according to claim 1, wherein the retaining member generally tapers in a longitudinal direction towards an end section of the insert and/or wherein the recess wall tapers in a depth direction of the recess.

7. The joinery arrangement according to claim 1, wherein respective ones of the projections extend, from a base surface of the retaining member, by different heights.

8. The joinery arrangement according to claim 1, wherein at least some of the projections extend, from a base surface of the retaining member, up to a virtual frustoconical surface, which forms a non-zero angle relative to the base surface.

9. The joinery arrangement according to claim 1, wherein, in the locked state, a locking portion of the insert cooperates with a locking surface of the recess and wherein there is a play between a head surface of a head portion of the insert and a recess wall arranged opposite to the cooperating recess wall of the recess.

10. The joinery arrangement according to claim 1, wherein the mounting side of the first joinery component is provided in a first face of the first joinery component and wherein the mounting area is provided in a first side edge of the second joinery component.

11. The joinery arrangement according to claim 1, wherein, during locking, the insert is configured to be compressed such that it assumes a compressed state, and wherein, subsequently, the insert is at least partially expanded towards an expanded state for locking the second joinery component to the first joinery component.

12. The joinery arrangement according to claim 1, wherein the second joinery component is configured to be locked to the first joinery component by relatively displacing the second joinery component with respect to the first joinery component in a locking direction extending essentially in parallel with at least one of said cavity direction and said protrusion direction.

13. The joinery arrangement according to claim 1, wherein the insert is arranged in an insert groove of the second joinery component.

14. The joinery arrangement according to claim 13, wherein the insert and/or an insert groove in which the insert is arranged essentially extends along a normal direction of the second plane.

15. The joinery arrangement according to claim 1, wherein the retaining member comprises an outer and/or an inner retaining member, the outer and inner retaining member being configured to cooperate with the recess and the insert groove, respectively, in the locked state.

16. The joinery arrangement according to claim 13, wherein the at least one projection is arranged in a body portion of the insert, the body portion being configured to cooperate with a groove wall of the insert groove in the locked state.

17. The joinery arrangement according to claim 1, wherein the insert comprises a plurality of

joined serpentine elements.

18. The joinery arrangement according to claim 1, wherein the protrusion angle is smaller than the cavity angle in an unlocked state of the first and the second joinery components.

19. The joinery arrangement according to claim 1, wherein the first and second joinery components are furniture components, and wherein the joinery arrangement is configured to be included in a furniture assembly.

20. A joinery assembly comprising the joinery arrangement according to claim 1, further comprising at least one additional joinery component.
